"TerraTec Cinergy T²"

DVB-T Highspeed USB2.0 communication protocol specification

Version 1.3
14/10/2004

http://qanu.de/

contact@qanu.de

Table of Contents

1. Enpoint 1

Communication with EP1 enables the host to set tuner parameters, query the current tuner status, start and stop the highspeed data stream.

The first byte – hereinafter refered to as the 'command byte' – specifies the type of request, a short description follows.

0x01 Reserved for future use
0x02 Reserved for future use
0x03 Start or stop the highspeed data stream
0x04 Set tuner parameters
0x05 Query current tuner parameters and status
0x06 Start brute-force channel scan
0x07 Continue brute-force channel scan
0x08 Get recent RC events
0x09 Sleep mode control

1.1 Start or stop the highspeed data stream

This command starts or stops the highspeed transfer on EP2.

1.1.1 Request

Endpoint	0x01
Command byte	0x03
Packet length (including command byte)	2 bytes

USB Packet Layout:

Byte #	Valid values	Function
0	0x03	Command byte
1	0x00, 0x01	Parameter byte

Parameter byte:

B7	B6	B5	B4	B3	B2	B1	B0
unused							Start

Set B0 to 1 to start the MPEG2 stream transfer over the USB endpoint 2, use 0 to stop it.

1.1.2 Return

An empty 0-byte bulk packet.

1.2 Set tuner parameters

1.2.1 Request

Endpoint	0x01
Command byte	0x04
Packet length (including command byte)	9 bytes

USB Packet Layout:

Byte #	Valid values	Function
0	0x04	Command byte
1-4	any	Frequency in kHz, (Little Endian)
5	0x06, 0x07, 0x08	Bandwidth in MHz
6-7	any	TPS parameters (Little Endian), see appendix A
8	any	Flags

Frequency specifies the requested Tuner Frequency, Bandwidth the transmission bandwidth. In Europe this is usually 7MHz for VHF and 8MHz for UHF channels.

If you don't force special parameters in the Flags byte, you can pass any value to the TPS word. For a detailled explanation of the TPS parameters see the ETSI DVB Specification.

Flags byte:

B7	B6	B5	B4	B3	B2	B1	B0
unused				force_ guard	force_ mode	force_ spec_inv	spec_inv

If force bits are not set, the demodulator probes the matching parameters automatically. In this case it might take a little more time to aquire a lock compared to a fully specified parameter set.

1.2.2 Return

A 0-byte bulk packet.

1.3 Query tuner parameters and status

1.3.1 Request

Endpoint	0x01
Command byte	0x05
Packet length (including command byte)	1 byte

This request returns a 25 byte USB packet indicating the tuner's current parameter settings and status bits. It does not take any parameter, the USB packet consists of the command byte only.

1.3.2 Return

USB Packet Layout:

Byte #	Function
0-3	Frequency in kHz (Little Endian)
4	Bandwidth in MHz
5-6	TPS parameters (Little Endian), see appendix A
7	Flags
8-9	Gain (Little Endian)
10	Signal-to-noise ration in dB
11-14	Viterbi bit error rate (Little Endian)
15-18	Reed Solomon bit error count (Little Endian)
19-22	Uncorrectable block count (Little Endian)
23-24	Lock/status indicator bits

Frequency, Bandwidth and TPS fields have the same meaning like the matching "Set Parameters" call.

Byte 7 - Flags:

B7	B6	B5	B4	B3	B2	B1	B0
unused							spec_inv

Bytes 8-9 reflect the AGC gain value which can be used as indicator of the incoming signal strength.

Bytes 11-14 contain the Viterbi Bit Error Rate. This is a good measurement of the signal quality and should be used in favor of the SNR.

Byte 19-22 contain the number of uncorrectable MPEG packets occurred since the last read operation.

Byte 23 – Lock/status indicator bits:

B7	B6	B5	B4	B3	B2	B1	B0
TPS_valid	BA_lock	FEC_lock	OFDM_ found	PILOT_ lock	DSCR_ lock	SYM_lock	AGC_lock

Description:

TPS_valid	Set to 0 if the TPS word is invalid at the end of a frame, e.g.due to a bad check sum.
BA_lock	Set when the Byte Aligner is in lock.
FEC_lock	Set when the forward error correction unit is working stable.
OFDM_found	Set when all OFDM pilots are found. Used to test for the presence of a DVB-T channel.
PILOT_lock	Set when OFDM continuous pilots are successfully received.
DSCR_lock	Set when the Descrambler is in lock.
SYM_lock	Set when the initial symbol timing lock has been established.

AGC_lock Set when the AGC is in lock.

Byte 24 - Lock/status indicator bits (continued):

B7	B6	B5	B4	B3	B2	B1	B0
unused							prev_ FEC_lock

prev_FEC_lock Set when a FEC lock has been achieved since the last read. This is interesting during a channel scan. If this bit is set, a channel has been found on the current frequency. The 'continue scan' command (0x07) needs to be issued to continue the scan. Please note that if there is no FEC_lock bit set, there has been a former lock which has gone away again, that usually indicates a bad reception quality.

Also, this bit is only set in scan mode.

For more detailed information about the meaning of the bits described here, please refer to Zarlink's MT352 design manual and to the ETSI DVB specification.

1.4 Start brute-force channel scan

The demodulator is capable of doing a brute-force channel scan over the entire possible frequency range.

Such a scan is initiated by the start command (command byte 0x06). After that, you need to poll the device's status (command byte 0x05). Once a channel is found (i.e. lock bits are set), you can continue the scan with command 0x07.

1.4.1 Request

Endpoint	0x01
Command byte	0x06
Packet length (including command byte)	10 bytes

USB Packet Layout:

Byte #	Valid values	Function
0	0x06	Command byte
1-4	any	start frequency in kHz (Little Endian)
5-8	any	end frequency in kHz (Little Endian)
9	0x06, 0x07, 0x08	Bandwidth in MHz

1.4.2 Return

A 0-byte bulk packet.

1.5 Continue brute-force channel scan

1.5.1 Request

Endpoint	0x01
Command byte	0x07
Packet length (including command byte)	1 byte

This request returns a 0-byte packet. It does not take any arguments.

1.5.2 Return

A 0-byte packet.

1.6 Get recent RC events

1.6.1 Request

Endpoint	0x01
Command byte	0x08
Packet length (including command byte)	1 byte

This request is used to read the latest RC events received by the receiver device.

1.6.2 Return

The firmware has a ring-buffer to store up to 8 decoded RC commands. If available, multiple events will be transmitted to the host in the return packet.
Each event block is 5 bytes long and has the following layout:

Byte #	Valid values	Function
0	0x01, 0x02	RC type - 0x01 for NEC, 0x02 for RC5 coding
1-4	any	32bit value of decoded RC command (Little Endian)

If no event is available, a 0-byte packet is transmitted, otherwise, the return packet has size of multiples of 5 bytes.
Please note that this endpoint has a size limit of 64 bytes, so a maximum of 12 events can be delivered at once.

1.7 Control sleep mode

1.7.1 Request

Endpoint	0x01
Command byte	0x09
Packet length (including command byte)	2 bytes

This command is used to control internal voltages on the Termini board. For reasons of power-saving, it is recommended to put the device to sleep whenever it is unused.

USB Packet Layout:

Byte #	Valid values	Function
0	0x09	Command byte
1	0x0, 0x1	Set to 0x1 to enter sleep mode, 0x0 for returning to normal operation.

This request returns a 0-byte packet. It does not take any argument.

1.7.2 Return

A 0-byte packet.

1.8 Get firmware version and sleep mode state

1.8.1 Request

Endpoint	0x01
Command byte	0x08
Packet length (including command byte)	1 byte

This request is used to read the latest RC events received by the Termini device.

1.8.2 Return

A 3-byte bulk packet is returned:

Byte #	Valid values	Function
0-1	any	Firmware version (Little Endian)
2	0x0, 0x1	Sleep mode state

2. Enpoint 2 (highspeed)

Once the highspeed stream has been started as described in section 1.1, bulk messages with 512 bytes each will get issued continously from EP2.

Buffers are not aligned to MPEG packet boundaries. All parts of the MPEG packets are transmitted, including the start byte (0x47).

Please note that the host side needs to provide a sufficient number of input buffers (USB request blocks) to catch all data packets and ensure a smooth transmission. At an average bandwidth of 14,5 Mbit/s, about 5000-6000 buffers arrive every second.

Appendix A – Transmission parameter signalling

The 16-bit TPS word describes the DVB-T transmission parameters and is assembled according to ETSI ETS 300744, section 4.6.2, table 9:

TPS	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0

<table>
<tr><th>function</th><th>HP/LP</th><th>const</th><th>hierarchy</th><th>HP code rate</th><th>LP code rate</th><th>guard</th><th>mode</th></tr>
<tr><td>000</td><td>HP</td><td>QPSK</td><td>None</td><td>1/2</td><td>1/2</td><td>1/32</td><td>2K</td></tr>
<tr><td>001</td><td>LP</td><td>QAM16</td><td>1</td><td>2/3</td><td>2/3</td><td>1/16</td><td>8K</td></tr>
<tr><td>010</td><td rowspan="4">n/a</td><td>QAM64</td><td>2</td><td>3/4</td><td>3/4</td><td>1/8</td><td rowspan="4">n/a</td></tr>
<tr><td>011</td><td rowspan="3">n/a</td><td>4</td><td>5/6</td><td>5/6</td><td>1/4</td></tr>
<tr><td>100</td><td rowspan="2">n/a</td><td>7/8</td><td>7/8</td><td rowspan="2">n/a</td></tr>
<tr><td>101-111</td><td>n/a</td><td>n/a</td></tr>
</table>